

Cosmological Hydrodynamics at Exascale: A Trillion-Particle Leap in Capability

Large Scale Hybrid Computing 25/26

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- Simulating the formation and evolution of the structure of the Universe is crucial for interpreting results of cosmological surveys.

N-body Simulations

- Simulate only gravity;
- Low fidelity for current surveys.

Cosmological Hydrodynamic Simulations

- Simulates gas dynamics, star formation, ...
- 10–20 $\times$ more computationally intensive.



- Frontier-E enables large-scale cosmological hydrodynamics through efficient exascale machine usage through CRK-HACC.

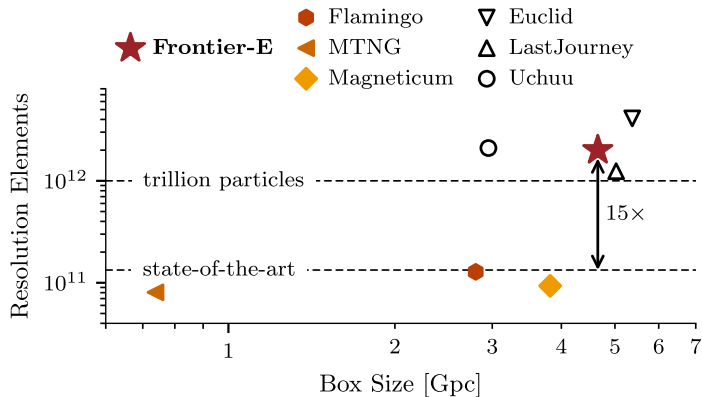


Figure 1. Box sizes and resolution elements of n-body (black) and hydrodynamic (colored) simulations

Frontier-E – Separation of Scale

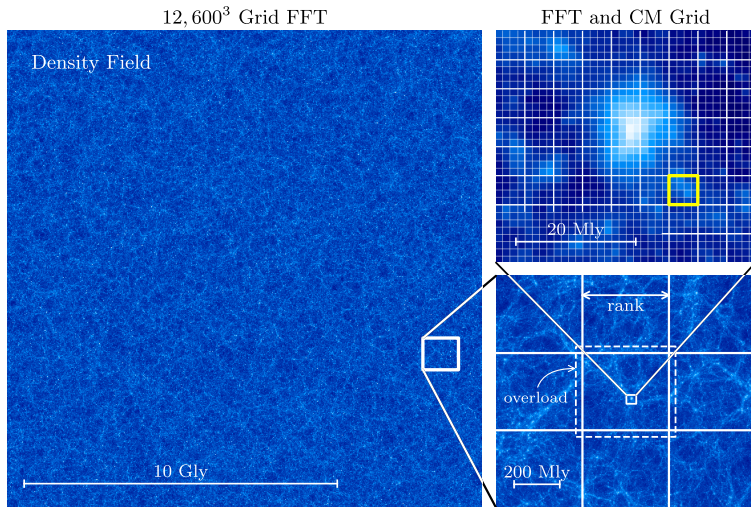


Figure 2. Domain subdivision and different interaction scales in Frontier-E



GPU Tree Solver

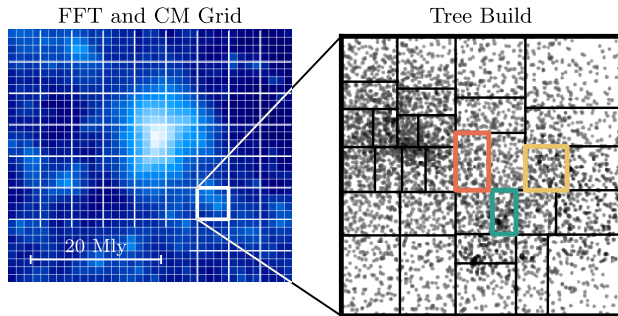


Figure 3. GPU Tree leaves generated for one CM bin.

Warp Splitting

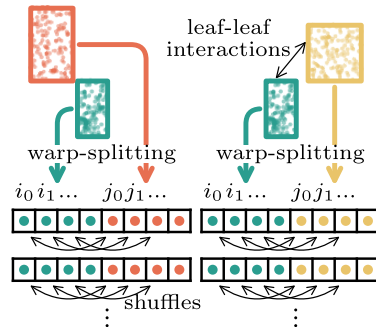


Figure 4. Short-range GPU solver.



In situ GPU-Accelerated End-to-End Analysis

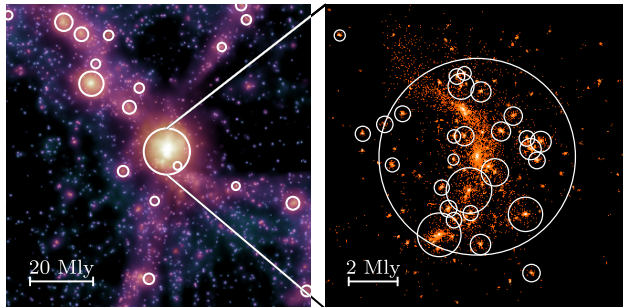


Figure 5. Halo (left) and galaxy (right) and detection

Multi-Tiered I/O

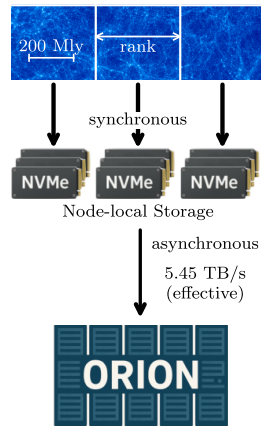


Figure 6. Frontier-E storage tiers

Frontier Supercomputer



Figure 7. Frontier (Oak Ridge National Laboratory)

128 – 9000 nodes



64-core AMD EPYC Zen 3 “Trento”



4× AMD Instinct MI250X



512 GiB DDR4 + 4× 64 GiB HBM2e



2× NVMe SSD (4 GB/s writes)



LustreFS PFS (4.6 TB/s writes)

Results – Parallel Scaling and Performance

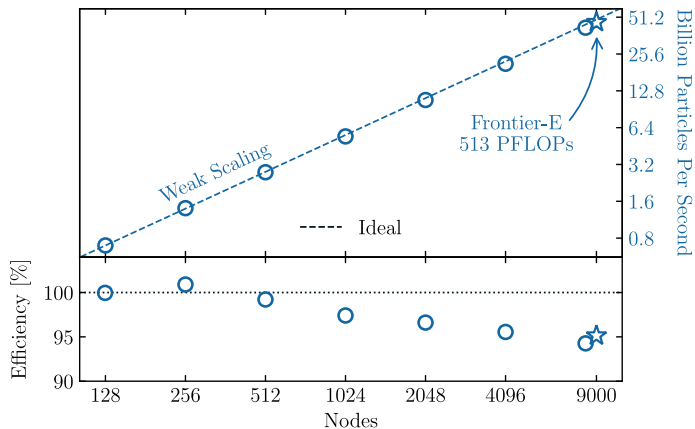
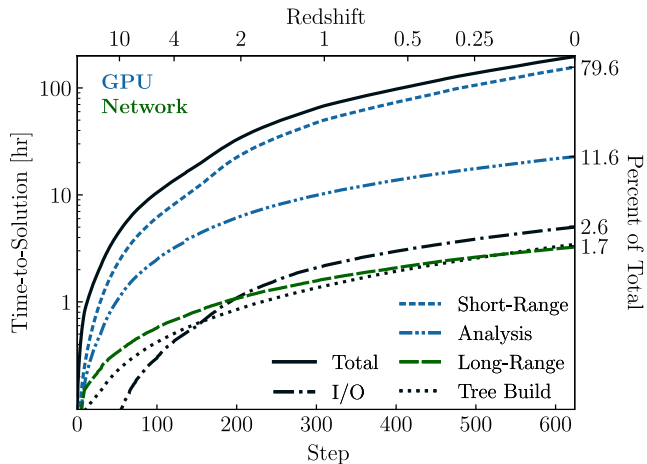


Figure 8. Weak scalability on Frontier

	PFLOPS	%
Sustained	420.5	24.4
Peak	513.1	29.8
Hardware Peak	1720.0	100.0

Table 1. Frontier supercomputer utilization

Results – Time to Solution



- 196 hours (8 days) of runtime;
- > 90% of time in GPU operations;

Figure 9. Cumulative time-to-solution of Frontier-E and its individual solvers

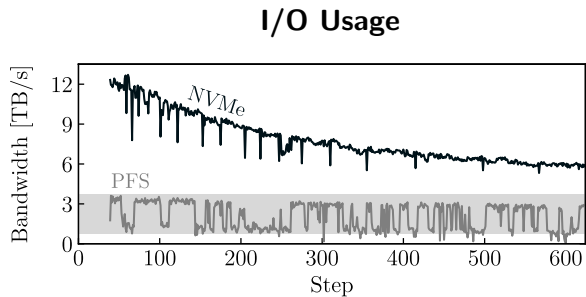


Figure 10. NVMe and PFS write bandwidth over time

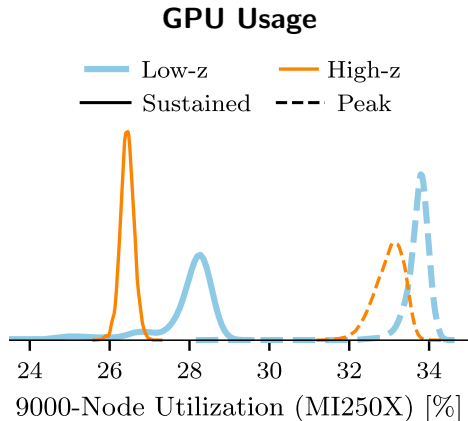


Figure 11. GPU utilization at different simulation stages



- Frontier-E provides the resolution and realism needed for next-generation surveys;
- Peak and sustained FLOP ratios of 513 and 420 PFLOPS were achieved;
- Techniques developed applicable to other simulations.

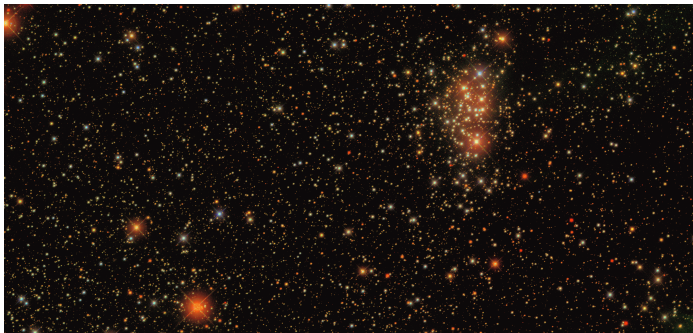


Figure 12. Sloan Digital Sky Survey (Robert Lupton)

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